Matthew Braaksma

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RESEARCH INTERESTS

Environment and Resource Economics · Land Use Economics · Ecosystem Services
Applied Econometrics · Causal Inference · Earth-Economy Modeling · Geospatial Computing

EDUCATION

University of Minnesota — Twin Cities, MN
Doctor of Philosophy: Applied Economics — Aug 2022 – Present

San Diego State University — San Diego, CA
Master of Arts: Economics — Aug 2020 – May 2022

Eastern Washington University — Cheney, WA
Bachelor of Arts: Business Administration, magna cum laude — Sept 2013 – June 2017
Minor: Economics, Visual Communication Design

SKILLS

Programming	Python, R, Stata, GAMS, Bash
Geospatial & Data	GDAL, geopandas, rasterio, xarray, pandas, pygeoprocessing, terra
Econometrics	Causal inference (difference-in-differences, fixed effects), discrete choice, hurdle models
Modeling	CGE modeling (CGEBox, MANAGE-WB), SEALS, InVEST, multiscale modeling
Computing	HPC (Minnesota Supercomputing Institute), Git, Quarto, Markdown, LaTeX

RESEARCH

Working Papers

Braaksma, Matthew. "Forest Protection and Landslide Regulation: A Global Ecosystem Service Methodology to Value Avoided Human Mortality". Under review at *Ecosystem Services*. SSRN. Code.

Dave, Dhaval, Yang Liang, Catherine Maclean, Joseph Sabia, and Matthew Braaksma. "Can Anti-Vaping Policies Curb Drinking Externalities? Evidence from E-Cigarette Taxation and Traffic Fatalities". No. w30670. *National Bureau of Economic Research*, 2022.

Publications

McWay, Ryan, and Matthew Braaksma. 2025. "The Political Consequences of Resource Scarcity: Targeted Spending in a Water-Stressed Democracy. A Replication Study of Mahadevan and Shenoy (Journal of Public Economics, 2023)". *World Development Perspectives* 39: 100707. https://doi.org/10.1016/j.wdp.2025.100707

Conference Presentations

"The Impacts of Extreme Weather on Human Mobility: A Minnesota Cooling Center Case Study" (with Dena Coffman, Ryan Harp, and Heidi Roop). Poster accepted, AGU Annual Meeting, San Francisco, CA, December 7–11, 2026.

"Global Spatially Explicit Land Supply Elasticity Estimates for Economic Modeling" (with Justin Johnson, Victor Nechifor, and Willem-Jan Van Zeist). 29th Annual Conference on Global Economic Analysis: Global Trade Analysis Project (GTAP), Kyoto, Japan, June 17–19, 2026.

"A Framework for Measuring African Ecosystem Services Impacts from Land-Related Policies" (with Justin Johnson, Victor Nechifor, Willem-Jan Van Zeist, Emanuele Ferrari, and Ole Boysen). 28th Annual Conference on Global Economic Analysis: Global Trade Analysis Project (GTAP), Kigali, Rwanda, June 25–27, 2025.

Professional Service

Abstract Reviewer, 29th Annual Conference on Global Economic Analysis, Global Trade Analysis Project (GTAP), Kyoto, Japan, June 2026.

Referee, *Southern Economic Journal*, 2025.

RESEARCH EXPERIENCE

University of Minnesota — Twin Cities, MN
Research Assistant, Supervised by Dr. Justin Johnson — June 2024 – Current

- *(June 2026 – Current)* Collaborate with The Nature Conservancy to link a CGE model with SEALS and InVEST for country-level analysis of conservation finance and commitments in the Amazon.
- *(September 2025 – Current)* Collaborated on development of K-SEALS, a fine-scale land use / land cover projection framework for Korea that translates Nature Futures scenario narratives into spatially explicit futures.
- *(June 2024 – December 2025)* Co-developed "A Framework for Measuring African Ecosystem Services Impacts from Land-Related Policies" with the European Commission Joint Research Centre and Wageningen University and Research, coupling a CGE model with InVEST and SEALS to evaluate land-use policy scenarios in Kenya.

NatCap TEEMs (Natural Capital Alliance: The Earth-Economy Modelers) — Twin Cities, MN
Student Member — August 2022 – Present

- Contributed landslide-regulation estimates to a multi-author global gross ecosystem product (GEP) study (in preparation).

University of Minnesota Climate Adaptation Partnership (MCAP) — Twin Cities, MN
Research Assistant, Supervised by Dr. Heidi Roop — June 2026 – August 2026

- Supported climate research applying geospatial computing to mobility and climate projection data to analyze cooling center accessibility under extreme heat.

TEACHING EXPERIENCE

Augsburg University — Minneapolis, MN
Adjunct Faculty — Fall 2026 – Current
ECO 112: Principles of Macroeconomics (course site)

University of Minnesota — Twin Cities, MN
Teaching Assistant

APEC 3002: Managerial Economics (Dr. Frances Homans)	Spring 2025
APEC 8213: Econometric Analysis III (Dr. Paul Glewwe)	Spring 2024, 2026
APEC 8214: Econometric Analysis IV (Dr. Marc Bellemare)	Spring 2024
APEC 8001: Consumer Choice and Demand (Dr. Paul Glewwe)	Fall 2023
APEC 8002: Production and Uncertainty (Dr. Rodney Smith)	Fall 2023

RECOGNITIONS

Abstract of Distinction, *UMN CFANS Research Symposium* — 2026
Teaching Assistant Award, 3000/4000-Level Classes, *UMN Applied Economics* — 2024–2025
Teaching Assistant Award, Graduate Classes, *UMN Applied Economics* — 2023–2024
Graduate Fellowship, *UMN Applied Economics* — 2022
Presidential Graduate Research Fellowship; McCuen Endowed Economics Scholarship, *SDSU* — 2020